Engineering Mathematics III

ENGE702

Lecture 11: Second Order Nonhomogeneous ODEs

Second order ODEs

From the table (in the last lecture), we know that our particular solution will be of the form

$$y_p = a\cos(\omega t) + b\sin(\omega t).$$

From here, we can take derivatives of y_p, and put it into our differential equation, and solve for a and b, and we get:

$$a = F_0 \frac{m(\omega_0^2 - \omega^2)}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2} \quad \text{and} \quad b = F_0 \frac{\omega c}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}$$

where

$$\omega_0 = \sqrt{\frac{k}{m}}$$

is called the 'natural frequency'.

Second order ODEs

Case 1: Undamped Forced Oscillations

Here $c = 0$. So in this case we have

$$a = \frac{F_0}{m(\omega_0^2 - \omega^2)} \quad \text{and} \quad b = 0$$

so our particular solution becomes

$$y_p = \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$$

Our general solution is then

$$y = A \sin(\omega_0 t) + B \cos(\omega_0 t) + \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$$

Second order ODEs

We will now look at what happens when $\omega = \omega_0$. In this case, one of the terms in our homogeneous solution y_h contains a $\cos(\omega_0 t)$ term, as does the suggested particular solution. So we need to use the modification rule, and our suggested particular solution becomes:

$$y_p = t(a\cos(\omega_0 t) + b\sin(\omega_0 t))$$

This gives (after finding the constants a and b):

$$y_p = \frac{F_0}{2m\omega_0} t\sin(\omega_0 t)$$

.

Second order ODEs

This gives displacements looking something like this:

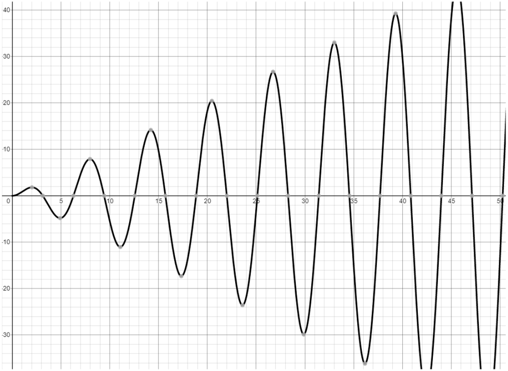


Figure: Resonance

In practical terms, this means the system would likely destroy itself, unless it caused it to be in a state where our model no longer applies.

Second order ODEs

Another similar phenomenon is known as **beats**. Here the driving frequency is 'close' to the natural frequency.

We showed earlier that the general solution will be

$$y = A\sin(\omega_0 t) + B\cos(\omega_0 t) + \frac{F_0}{m(\omega_0^2 - \omega^2)}\cos(\omega t)$$

Suppose we have a particular solution (for a particular set of initial conditions) that is

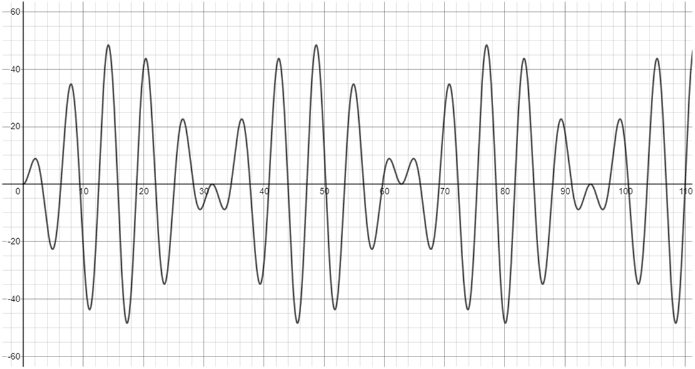
$$y = \frac{F_0}{m(\omega_0^2 - \omega^2)}\cos(\omega t) - \frac{F_0}{m(\omega_0^2 - \omega^2)}\cos(\omega_0 t)$$

Second order ODEs

This can be rewritten as:

$$y = \frac{2F_0}{m(\omega_0^2 - \omega^2)} \sin\left(\frac{\omega_0 + \omega}{2} t\right) \sin\left(\frac{\omega_0 - \omega}{2} t\right)$$

When ω_0 and ω are close, then the period of the last $\sin$ function is large. So we end up with something like:



Second order ODEs

Case 2: Damped Forced Oscillations

Again the solution is a combination of the homogeneous solution and a particular solution.

But recall from the section on unforced oscillation that the unforced damped system will always tend to 0 as time increases. This means that:

'After a sufficiently long time the output of a damped vibrating system under a purely sinusoidal driving force will practically be a harmonic oscillation whose frequency is that of the input.'

So for damped forced oscillations, there is the **transient phase**, where the homogeneous solution will be involved, ie $y = y_h + y_p$, and the **steady-state phase**, where the homogeneous solution is negligible, so $y = y_p$.

Second order ODEs

In the damped case, we do not have ‘run-away’ resonance. Oscillations will always be finite (even theoretically), but some form of resonance still exists. This is called **practical resonance**.

The first thing we will note is that $y_p(t) = a\cos(\omega t) + b\sin(\omega t)$ can be rewritten

$$y_p = C^* \cos(\omega t - \eta)$$

where C^* is called the **amplitude** and η is called the **phase angle**.

We can derive (see the textbook for details) that the **steady state amplitude** at a particular frequency ω is given by:

$$C^*(\omega) = \frac{F_0}{\sqrt{m^2\left(\omega_0^2 - \omega^2\right)^2 + \omega^2 c^2}}.$$

Second order ODEs

After some further derivation, we can find that the frequency at which the greatest amplitude occurs, called ω_{max}, is given by

$$\omega_{max}^2 = \omega_0^2 - \frac{c^2}{2m^2}$$

with a corresponding amplitude of

$$C^*(\omega_{max}) = \frac{2mF_0}{c\sqrt{4m^2\omega_0^2 - c^2}}.$$

Second order ODEs

Example: Find the frequency ω that would give the greatest amplitude of oscillation in a system modelled with the following differential equation

$$2y'' + 4y' + 7y = 4\cos(\omega t)$$

Second order ODEs

In this lecture we have so far dealt with a mechanical forced mass - spring - damper system. There is a completely analogous electrical system: that of a resistor - inductor - capacitor (RLC) circuit with an applied voltage.

If we have a resistance R, an inductance L, a capacitance C, and an EMF of $E_0 \sin(\omega t)$, the current in the circuit, I, is modelled by

$$LI'' + RI' + \frac{1}{C}I = E_0 \omega \cos(\omega t)$$

We will not go into the details of this in this lecture (it is covered in Chapter 2.9 in the textbook), but will note that there is a direct analogy between this model and the mechanical model we have studied today.

Exercises

- Find the steady state amplitude and the frequency of the systems modelled by the following differential equations:
  - $3y'' + 2y' + y = 5\cos(4t)$
  - $y'' + 3y' + 6y = 2\cos(13t)$
- What driving frequency will give the maximum amplitude of oscillation for each of the above systems, and what will be the amplitude at this frequency?

Text Book References

Chapter 2.8